BUDT703_Project_0507_06_Report
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1. **Data Processing**
We summarize the data processing steps undertaken to transform the original Amtrak dataset into its final processed form. The original dataset contained general information about Amtrak stations, while the data set we processed focused on ridership, OTP, AGR, procurement for each state. We have added a unique ID to each dataset to ensure that each record can be uniquely identified and to support cross-table analysis and queries.

Ridership:
This dataset consolidates ridership data by state and includes yearly ridership.

Columns:

- ridershipId: A unique identifier for each state or region.
- ridershipState: The name of the state.
- ridershipAmount_21: Ridership data for the year 2021.
- ridershipAmount_22: Ridership data for the year 2022.
- ridershipAmount_23: Ridership data for the year 2023.

On-Time Performance (OTP):

This dataset consolidates On-Time Performance (OTP) data by state and includes yearly OTP.

Columns:

- OTPId: A unique identifier for on-time performance data.
- OTPState: The name of the state or region.
- OTPAmount_21: On-time performance percentage for 2021.
- OTPAmount_22: On-time performance percentage for 2022.
- OTPAmount_23: On-time performance percentage for 2023.

Guest Rewards (AGR):

This dataset aggregates guest reward metrics by state for yearly analysis.

Columns:

- AGRId: A unique identifier for guest reward data.
- AGRState: The name of the state or region.
- AGRAmount_21: Guest rewards data for the year 2021.
- AGRAmount_22: Guest rewards data for the year 2022.
- AGRAmount_23: Guest rewards data for the year 2023.

Route:

The data for the route dataset is derived from the OTP dataset, but the route dataset focuses on the information for each route.

Columns:

- routeId: A unique identifier for each route.
- routeName: The name of the route.
- routeType: The type of route.
- routeStation: Associated station(s) for the route.
- routeHostRailroad: Host railroad information for the route.
- routeAmount_21: Performance or ridership metrics for 2021.
- routeAmount_22: Performance or ridership metrics for 2022.
- routeAmount_23: Performance or ridership metrics for 2023.

Procurement:

This dataset focuses on procurement amount yearly across states.

Columns:

- procurementId: A unique identifier for procurement data.
- procurementState: The name of the state or region.
- procurementAmount_21: Procurement expenditure on goods and services for 2021.
- procurementAmount_22: Procurement expenditure on goods and services for 2022.

- procurementAmount_23: Procurement expenditure on goods and services for 2023.

Correlate:

This dataset explores correlations between routes and ridership data.

Columns:

- routeId: A unique identifier for each route.
- ridershipId: A unique identifier for ridership data.
- routeStation: Route Station information to establish correlations between Route dataset and Ridership dataset.

Depend:

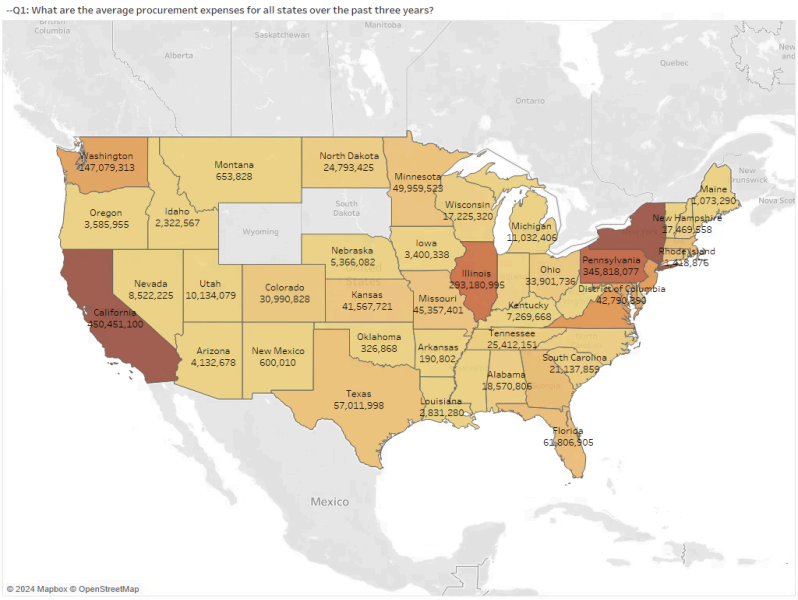
This dataset explores correlations between routes and OTP data.

Columns:

- routeId: A unique identifier for each route.
- OTPId: A unique identifier for on-time performance data.
- routeHostRailroad: Host Railroad information to establish correlations between Route dataset and OTP dataset.

2. **Insights**

a. What are the average procurement expenses for all states over the past three years?



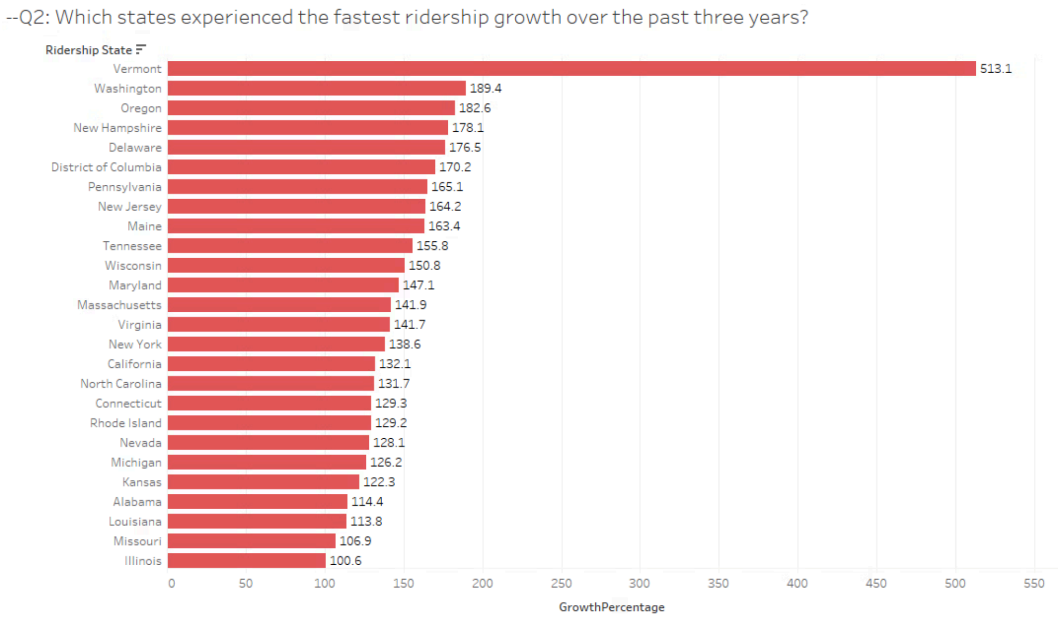
a.1. Optimize Budget Allocation for High-Spending States

Since California and New York states have the highest average procurement expenses, allocate a larger portion of the procurement budget to them. To do this effectively, we could break down procurement expenses into some categories—such as equipment, maintenance, and infrastructure—so we can target the high-spending states.

a.2. Assess Regional Analysis

Besides, we could assess regional analysis. As we can see in the table, after Pennsylvania and Illinois, Virginia has a significant drop in procurement expenses. This suggests that there may be potential differences in operational scale or efficiency. And also, there may be some regional factors such as supplier availability and labor costs. For example, California and New York state might have higher material and labor costs. So, to analyze effectively, we could determine whether high-spending states are paying more for similar services than lower-cost states.

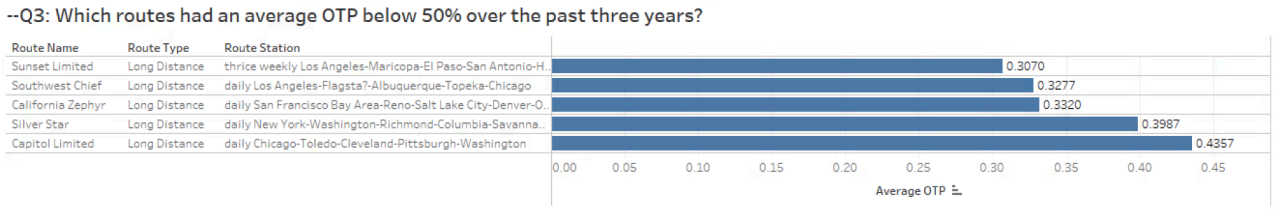
b. Which states experienced the fastest ridership growth over the past three years?



b.1. Vermont Leads in Growth Percentage

Vermont experienced an extraordinary 513.1% growth in ridership, making it the state with the fastest growth rate. This significant increase indicates an emerging demand for transportation services or improved infrastructure over the past three years.

c. Which routes had an average OTP below 50% over the past three years?



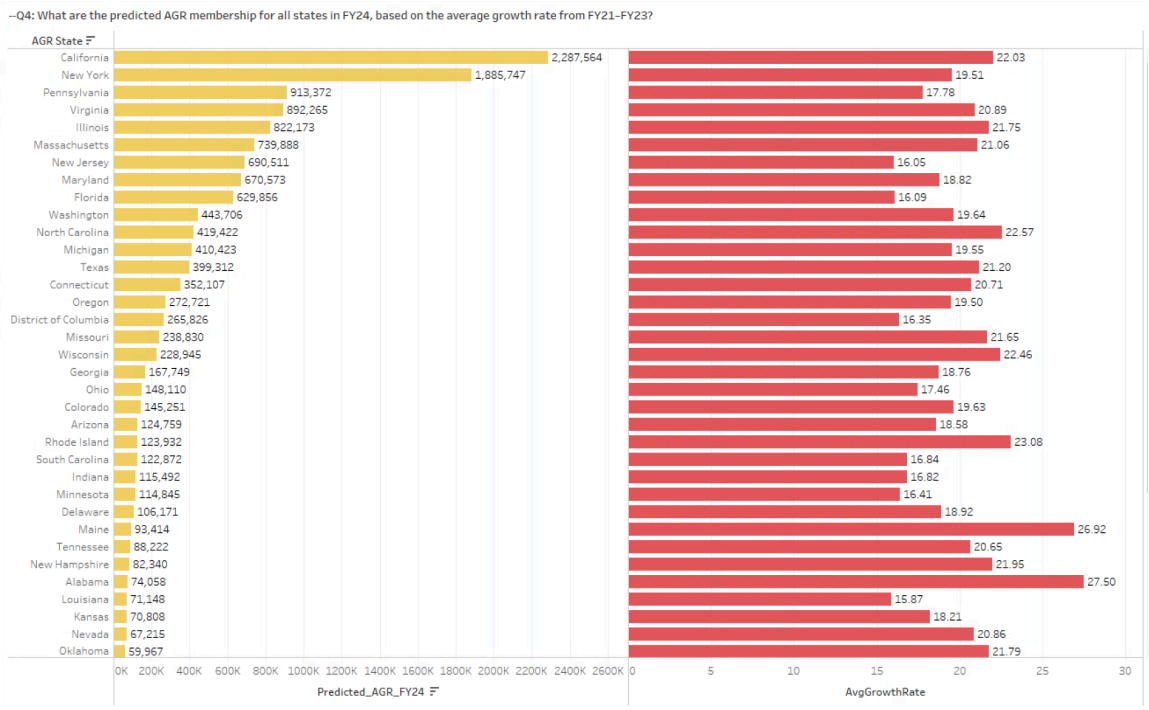
c.1 Delays in Specific Routes

The Sunset Limited(30.70%) and Southwest Chief(32.77%) routes show the lowest On-Time Performance(OTP), reflecting significant operational challenges in meeting schedules and inefficiencies in long-distance train services.

c.2 Operational and External Factors

The low OTP performance can be attributed to factors such as the vast geographical coverage of these routes, the complexity of managing long-distance operations, and external challenges such as station congestion. For instance, routes like "California Zephyr" (33.20%) and "Silver Star" (39.87%) likely face delays due to the logistical difficulties in multi-state operations.

d. What are the predicted AGR membership for all states in FY24, based on the average growth rate from FY21–FY23?



d.1. Dominance of California

California leads the list by a wide margin due to its large AGR membership base and a high average growth rate (22.03%).

d.2. Slightly Slower Growth States

New Jersey and Florida have relatively slower growth rates but maintain their positions due to a stable AGR membership base.

3. **Conclusion**

a. East Coast States(e.g., New York and Pennsylvania) dominate in procurement expenses and exhibit steady AGR membership growth.

b. Washington performs well in procurement but shows relatively lower AGR membership growth potential, possibly due to smaller market size or intense competition.

c. While Vermont leads the nation with an impressive 513.1% ridership growth, indicating rising demand and potential infrastructure improvements, long-distance

train services face significant challenges in maintaining on-time performance and reliability. These insights highlight the need to address operational inefficiencies while capitalizing on emerging growth opportunities.

4. **Recommendations**

a. In east coast markets(e.g., New York, New Jersey, Pennsylvania), we could develop high-value membership programs for high population density and busy transportation networks. Besides, we would optimize the procurement network, reduce logistics costs, and improve transportation reliability.

b. In west coast markets(e.g., California, Washington), for the large member base, we could enhance high-end services to attract high-income groups. Additionally, exploring sustainable solutions in sourcing and collaborating with environmentally friendly suppliers could enhance brand image.

c. Use the success in Vermont as a model to replicate improvements in other high-growth areas while addressing systemic challenges in long-distance services.

d. Reallocate resources strategically to both growth states and underperforming routes to balance service quality and expansion.

5. **References**

https://www.amtrak.com/about-amtrak/amtrak-facts/state-fact-sheets.html
https://www.amtrak.com/about-amtrak/reports-documents.html
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